

Interfaces Configuration of 5 nodes

eth0

Speed: N/A

Admin (PXE)

Offloading Modes: [Default](#) MTU: [Default](#)

eth1

Speed: N/A

Management

Offloading Modes: [Default](#) MTU: [Default](#)

eth2

Speed: N/A

Storage

Offloading Modes: [Default](#) MTU: [Default](#)

eth3

Speed: N/A

Public

Figure 9: Assigning networks

```
br-ex      8000.1655a9613cc9    no    eth3
          p_ff798dba-0
          v_public
          v_router_pub

br-fw-admin 8000.525400daba79    no    eth0
br-mesh     8000.52540027a872    no    eth4
br-mgmt     8000.5254009b320e    no    eth1
          mgmt-conntrd
          v_router

br-storage  8000.525400ec6d44    no    eth2
```

Advanced configuration of Fuel

On a large environment with hundreds of servers spread across multiple racks, a single broadcast domain can cause performance and maintenance issues. In order to address this, Fuel has the *networknodegroup* concept, where each rack is organised into one *nodegroup* with different network subnets and gateways.

The following is an environment with seven racks of servers:

Figure 10: Deploying changes

Figure 11: Successful deployment

to list *nodegroups* from cli

```
fuel nodegroup
id | cluster_id | name
-----
1 | 1 | default
2 | 1 | rack10
3 | 1 | rack11
4 | 1 | rack12
5 | 1 | rack13
6 | 1 | rack14
7 | 1 | rack15
```

Distributing controller roles

In this set-up, I have used three servers to run all the controller services; but it is possible to have separate Keystone, Rabbitmq, MySQL, MongoDB and HAProxy services roles assigned to different servers by installing additional Fuel plugins. In a large environment, it is recommended to have separate controllers for each service, for better performance. **END**

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